Join our community, grow your skills, and stay up to date with Epic's latest news and updates.
Sign In or Create an Account Dismiss

DOCUMENTATION
Unreal Engine
Home
Documentation
Unreal Engine
Release Notes
Manuals 5.0 Reference
C++ API Reference
Learning Paths
Learning
Courses Assessment
Images
Videos
Tutorials
Unreal Engine 5.0.3
Filter by title
Unreal Engine 5.0 Documentation
What's New
Understanding the Basics
Mobile Development
Working with Content
Building Worlds
Engaging Heads, Bodies, and Faces
Creating Visual Effects
Cinematics and Movies
Animation Tools
Artificial Intelligence
Blueprint Scripting Reference
Networking, Multiplayer, and Dedicated Servers
Gameplay Ability System
Programming Multiplayer Games
Networking Reference
Gameplay Framework
Programming Multiplayer Games
Online Beacon Host
Engine Example
Process Flow of Test Beacons
Engine Configuration
Configuration Reference
Online Beacons
Testing and Debugging
Setting up the Mobile Platform
Optimizing Your Content
Staying and Networking Projects
Unreal Engine Source and Contribution
Unreal Engine Glossary of Terms
Unreal Engine Blueprint API Reference
Unreal Engine C++ API Reference
Unreal Engine Console Variables and Commands
Glossary
Unreal Engine Extension and Dispatch Reference
Creating User Interfaces
Animation Tools
Artificial Intelligence
Working with Media
Setting up the

```
APingHost::APingHost(const FObjectInitializer& ObjectInitializer) : Super(ObjectInitiali
{
  BeaconState = EBeaconState::AllowRequests;
}

bool APingHost::InitializeHost()
{
  bInitSuccess = InitHost();
  return bInitSuccess;
}

void APingHost::RegisterHostObject(AOnlineBeaconHostObject* HostObject)
{
  if (bInitSuccess)
  {
    RegisterHost(HostObject);
  }
}
```

Copy full snippet

Ping Host Object

PingHostObject.h

```
#pragma once

#include "CoreMinimal.h"
#include "OnlineBeaconHostObject.h"
#include "PingHostObject.generated.h"

DECLARE_LOG_CATEGORY_EXTERN(LogPingHostObject, Log, All);

/**
*
*/
UCLASS(Blueprintable, BlueprintType, Transient, NotPlaceable, Config = Engine)
class BEACONS_API APingHostObject : public AOnlineBeaconHostObject
{
  GENERATED_BODY()

public:
  APingHostObject(const FObjectInitializer& ObjectInitializer);

  virtual AOnlineBeaconClient* SpawnBeaconActor(class UNetConnection* ClientConnection)

  virtual void OnClientConnected(class AOnlineBeaconClient* ClientActor, class UNetConne

  virtual bool Init();
};
```

Copy full snippet

PingHostObject.cpp

```
#include "PingHostObject.h"
#include "PingClient.h"

DEFINE_LOG_CATEGORY(LogPingHostObject)

APingHostObject::APingHostObject(const FObjectInitializer& ObjectInitializer) : Super(OI
{
  ClientBeaconActorClass = APingClient::StaticClass();
  BeaconTypeName = ClientBeaconActorClass->GetName();

  PrimaryActorTick.bCanEverTick = true;
  PrimaryActorTick.bAllowTickOnDedicatedServer = true;
  PrimaryActorTick.bStartWithTickEnabled = true;
}

AOnlineBeaconClient* APingHostObject::SpawnBeaconActor(UNetConnection* ClientConnection)
{
  return Super::SpawnBeaconActor(ClientConnection);
}

void APingHostObject::OnClientConnected(AOnlineBeaconClient* ClientActor, UNetConnectior
{
  Super::OnClientConnected(ClientActor, ClientConnection);
  APingClient* BeaconClient = Cast<APingClient>(ClientActor);
  if (BeaconClient != nullptr)
  {
    BeaconClient->ClientPingBegin();
  }
}

bool APingHostObject::Init()
{
  return true;
}
```

Copy full snippet

Game Mode Code

BeaconsGameMode.h

```
#pragma once

#include "CoreMinimal.h"
#include "GameFramework/GameModeBase.h"
#include "BeaconsGameMode.generated.h"

UCLASS(minimalapi)
class ABeaconsGameMode : public AGameModeBase
{

  GENERATED_BODY()

public:
  ABeaconsGameMode();

  bool InitPingBeacon();

protected:
  virtual void BeginPlay();
};
```

Copy full snippet

BeaconsGameMode.cpp

```
// Copyright Epic Games, Inc. All Rights Reserved.

#include "BeaconsGameMode.h"
#include "BeaconsCharacter.h"
#include "UObject/ConstructorHelpers.h"
#include "PingBeacon/PingHost.h"
#include "PingBeacon/PingHostObject.h"

ABeaconsGameMode::ABeaconsGameMode()
{
  // set default pawn class to our Blueprinted character
  static ConstructorHelpers::FClassFinder<APawn> PlayerPawnBPClass(TEXT("/Game/ThirdPers
  if (PlayerPawnBPClass.Class != NULL)
  {
    DefaultPawnClass = PlayerPawnBPClass.Class;
  }
}

bool ABeaconsGameMode::InitPingBeacon()
{
  ENetMode Mode = GetNetMode();
  if (Mode == NM_DedicatedServer || Mode == NM_ListenServer)
  {
    APingHost* Host = GetWorld()->SpawnActor<APingHost>(APingHost::StaticClass());
    if (Host && Host->InitializeHost())
    {
      APingHostObject* HostObject = GetWorld()->SpawnActor<APingHostObject>(APingHostObj
      if (HostObject)
      {
        Host->RegisterHostObject(HostObject);
        return true;
      }
    }
  }
  return false;
}

void ABeaconsGameMode::BeginPlay()
{
  bool bSuccess = InitPingBeacon();
  if (bSuccess)
  {
    UE_LOG(LogTemp, Log, TEXT("PingBeacon Init Success"))
  }
  else
  {
    UE_LOG(LogTemp, Log, TEXT("PingBeacon Init Failure"))
  }
}
```

Copy full snippet

Character Blueprint

Copy code

gameplay multiplayer network advanced

Ask questions and help your peers Developer Forums

Write your own tutorials or read those from others Learning Library

Collapse